

# Exercise Sheet 4

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## Naive Bayes and the Generative Route

Probabilistic Machine Learning

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# 4.A The Generative Route

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# Exercise 4.A.1: Counting Parameters

A fraud classifier (two classes) uses four attributes: **payment** (card, invoice, wallet), **country** (domestic, EU, other), **new device** (yes/no) and **night** (yes/no).

1. How many free numbers does the full generative model need — the class prior and one joint  $p(x \mid y)$  per class?
2. How many does naive Bayes need?
3. For  $p$  yes/no attributes and  $K$  classes, give both counts as formulas.
4. From which  $p$  on does the full joint need more than a million numbers *per class*?

# Exercise 4.A.1: Counting Parameters — Solution

# Exercise 4.A.1: Counting Parameters — Solution

- 1. Full model.** Each class needs a table over  $3 \cdot 3 \cdot 2 \cdot 2 = 36$  value combinations, of which 35 are free (they sum to 1). Two classes give 70; the prior  $P(\text{fraud})$  adds one: **71**.
- 2. Naive Bayes.** Per class one categorical per attribute:  $(3 - 1) + (3 - 1) + (2 - 1) + (2 - 1) = 6$  free numbers. Two classes give 12, plus the prior: **13**.
- 3. Formulas.** Full:  $K(2^p - 1) + (K - 1)$ . Naive Bayes:  $Kp + (K - 1)$ .
- 4.**  $2^{19} - 1 = 524,287$  and  $2^{20} - 1 = 1,048,575$ , so from  $p = 20$  on. The naive count grows linearly: 20 per class.

## Exercise 4.A.2: Fitting Is Counting

An attribute takes  $k$  values. Within one class we observed it  $n$  times, value  $v$  exactly  $n_v$  times ( $\sum_v n_v = n$ ). Model it as categorical with parameters  $\theta_1, \dots, \theta_k \geq 0, \sum_v \theta_v = 1$ .

1. Write down the log-likelihood  $\ell(\theta)$  of the  $n$  i.i.d. observations.
2. For  $k = 2$ , substitute  $\theta_2 = 1 - \theta_1$  and show that the MLE is  $\hat{\theta}_1 = n_1/n$  — the  $h/n$  of Lecture 03.
3. For general  $k$ , maximise  $\ell$  under the constraint  $\sum_v \theta_v = 1$  (Lagrange multiplier) and show  $\hat{\theta}_v = n_v/n$ .
4. Why can naive Bayes fit every attribute's table separately, while a full joint  $p(x \mid y)$  could not be fitted attribute by attribute?

# Exercise 4.A.2: Fitting Is Counting — Solution

## Exercise 4.A.2: Fitting Is Counting — Solution

1. Each observation of value  $v$  contributes  $\theta_v$ :

$$\ell(\theta) = \sum_{v=1}^k n_v \log \theta_v$$

2.  $\ell = n_1 \log \theta_1 + n_2 \log(1 - \theta_1)$ . Setting  $\ell'(\theta_1) = n_1/\theta_1 - n_2/(1 - \theta_1) = 0$  gives  $n_1(1 - \theta_1) = n_2\theta_1$ , so  $\hat{\theta}_1 = n_1/(n_1 + n_2) = n_1/n$ .

3.  $L(\theta, \lambda) = \sum_v n_v \log \theta_v - \lambda(\sum_v \theta_v - 1)$ . From  $\partial L/\partial \theta_v = n_v/\theta_v - \lambda = 0$  we get  $\theta_v = n_v/\lambda$ ; the constraint gives  $\lambda = \sum_v n_v = n$ , so  $\hat{\theta}_v = n_v/n$  — the relative frequency.

4. Under the naive assumption the log-likelihood of the table is a **sum** of one term  $\sum_v n(x_j = v, y) \log \theta_{jvy}$  per attribute and class; no parameter appears in two terms, so each term is maximised on its own — by counting. A full joint has one parameter per *combination* of values; its counts are counts of whole rows, and they cannot be split into per-attribute counts.

# 4.B Naive Bayes by Hand

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## Exercise 4.B.1: Twelve Transactions

Flags: new device  $D$ , foreign IP  $F$ , night  $N$ , high amount  $H$  (1 = yes).

| # | $D$ | $F$ | $N$ | $H$ | fraud? | #  | $D$ | $F$ | $N$ | $H$ | fraud? |
|---|-----|-----|-----|-----|--------|----|-----|-----|-----|-----|--------|
| 1 | 1   | 1   | 1   | 1   | yes    | 7  | 1   | 0   | 0   | 0   | no     |
| 2 | 1   | 1   | 0   | 1   | yes    | 8  | 0   | 0   | 0   | 0   | no     |
| 3 | 1   | 0   | 1   | 0   | yes    | 9  | 0   | 1   | 0   | 0   | no     |
| 4 | 0   | 1   | 1   | 1   | yes    | 10 | 0   | 0   | 0   | 0   | no     |
| 5 | 0   | 0   | 0   | 0   | no     | 11 | 1   | 0   | 1   | 0   | no     |
| 6 | 0   | 0   | 1   | 0   | no     | 12 | 0   | 0   | 0   | 0   | no     |

1. Estimate  $P(\text{fraud})$  and  $P(\text{flag} = 1 \mid \text{class})$  for all four flags by counting.
2. Compute  $P(\text{fraud} \mid x)$  for  $x_1 = (D, F, N, H) = (1, 0, 0, 0)$ .
3. Do the same for  $x_2 = (0, 0, 0, 1)$ . What goes wrong, and why?
4. Repeat 2 and 3 with Laplace smoothing on the flags (keep the prior unsmoothed). Which result changes a lot, which only a little?

# Exercise 4.B.1: Twelve Transactions — Solution

## Exercise 4.B.1: Twelve Transactions — Solution

1.  $P(\text{fraud}) = 4/12 = 1/3$ . Among the 4 frauds, each flag occurs 3 times:  $3/4$  for  $D, F, N, H$ . Among the 8 honest:  $D\ 2/8, F\ 1/8, N\ 2/8, H\ 0/8$ .

2. Fraud:  $\frac{1}{3} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{256} \approx 0.0039$ . Honest:  $\frac{2}{3} \cdot \frac{2}{8} \cdot \frac{7}{8} \cdot \frac{6}{8} \cdot 1 = 0.1094$ .  $P(\text{fraud} \mid x_1) = 0.0039 / (0.0039 + 0.1094) = 1/29 \approx 0.034$ .

3. Fraud:  $\frac{1}{3} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{1}{256}$ . Honest:  $\dots \cdot P(H=1 \mid \text{honest}) = 0$ . So  $P(\text{fraud} \mid x_2) = 1$ : a transaction without any red flag except a high amount is **certain** fraud, because no honest high amount occurred in eight rows. One zero count vetoes the other three flags, which all speak for honest.

4. Smoothed,  $(\text{count} + 1)/(n + 2)$ : fraud  $4/6 = 2/3$  for every flag; honest  $D\ 0.3, F\ 0.2, N\ 0.3, H\ 0.1$ .  $x_1$ : fraud  $\frac{1}{3} \cdot \frac{2}{3} \cdot (\frac{1}{3})^3 = 0.0082$ , honest  $\frac{2}{3} \cdot 0.3 \cdot 0.8 \cdot 0.7 \cdot 0.9 = 0.1008$ , so  $P = 0.075$  (from 0.034).  $x_2$ : fraud 0.0082, honest  $\frac{2}{3} \cdot 0.7 \cdot 0.8 \cdot 0.7 \cdot 0.1 = 0.0261$ , so  $P \approx 0.24$  (from 1). With 4 and 8 rows the prior of one pseudo-count per value is not small, so even  $x_1$  moves — smoothing matters most where the counts are smallest.

## Exercise 4.B.2: Smoothing Is a Prior

A yes/no flag was seen  $h$  times among  $n$  frauds.

1. With a  $\text{Beta}(1, 1)$  prior on  $\theta = P(\text{flag} \mid \text{fraud})$ , what is the posterior (Lecture 02)? Show that its **mean** is Laplace's  $(h + 1)/(n + 2)$ .
2. What is the posterior's **peak** for  $h = 0$ ? What does that say about "keep the peak"?
3. "Add- $\alpha$ " smoothing uses  $(h + \alpha)/(n + 2\alpha)$ . Which prior belongs to it? Compute the estimate for  $h = 0, n = 100$  and  $\alpha = 0.5, 1, 10$ .
4. A merchant category with  $k = 10$  values; one category never occurred in 100 frauds. What does Laplace smoothing give, and which prior is behind it?

# Exercise 4.B.2: Smoothing Is a Prior — Solution

## Exercise 4.B.2: Smoothing Is a Prior — Solution

1. Adding counts:  $\text{Beta}(1 + h, 1 + n - h)$ , with mean  $\frac{1+h}{(1+h)+(1+n-h)} = \frac{h+1}{n+2}$ .
2. The mode of  $\text{Beta}(a, b)$  is  $\frac{a-1}{a+b-2}$ ; here  $\frac{h}{n} = 0$ . Under a flat prior the peak is the MLE, so keeping the peak keeps the zero. Laplace smoothing silently switches to the mean.
3.  $\text{Beta}(\alpha, \alpha)$ : posterior  $\text{Beta}(\alpha + h, \alpha + n - h)$ , mean  $(h + \alpha)/(n + 2\alpha)$ . For  $h = 0$ ,  $n = 100$ :  $\alpha = 0.5$ :  $0.5/101 \approx 0.0050$ ;  $\alpha = 1$ :  $1/102 \approx 0.0098$ ;  $\alpha = 10$ :  $10/120 \approx 0.083$ . A larger  $\alpha$  is a stronger belief that  $\theta$  is near  $1/2$  — worth  $2\alpha$  imagined observations.
4.  $(0 + 1)/(100 + 10) = 1/110 \approx 0.0091$  — the posterior mean under a flat **Dirichlet**  $(1, \dots, 1)$ , the Beta for more than two values.

# 4.C Generative vs Discriminative

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## Exercise 4.C.1: The Same Line

Naive Bayes with yes/no flags,  $\theta_{jy} = P(x_j = 1 \mid y)$ , prior  $\pi = P(y = 1)$ . Lecture numbers:  $\pi = 1/1000$ ; new device 0.6 vs 0.1; foreign IP 0.7 vs 0.05; night 0.5 vs 0.2 (fraud vs honest).

1. Show that  $\log \frac{P(y=1|x)}{P(y=0|x)} = w_0 + \sum_j w_j x_j$  and give  $w_j$  and  $w_0$ .
2. Compute the three weights and  $w_0$ , and check that a transaction with all three flags gets  $P(\text{fraud} \mid x) \approx 17.4\%$ .
3. The foreign-IP flag is **missing**. Compare what naive Bayes should do with plugging  $x_{\text{IP}} = 0$  into the linear form.
4. Why can logistic regression, which has the same linear form, not do what naive Bayes does in 3?

## Exercise 4.C.1: The Same Line — Solution

## Exercise 4.C.1: The Same Line — Solution

1.  $\log p(x_j \mid y) = x_j \log \theta_{jy} + (1 - x_j) \log(1 - \theta_{jy})$ . Take the difference between the classes and sum over  $j$ :

$$w_j = \log \frac{\theta_{j1}}{\theta_{j0}} - \log \frac{1 - \theta_{j1}}{1 - \theta_{j0}}, \quad w_0 = \log \frac{\pi}{1 - \pi} + \sum_j \log \frac{1 - \theta_{j1}}{1 - \theta_{j0}}$$

2.  $w_{\text{dev}} = \log 6 - \log \frac{0.4}{0.9} = 1.792 + 0.811 = 2.603$ ;  $w_{\text{IP}} = \log 14 - \log \frac{0.3}{0.95} = 2.639 + 1.153 = 3.792$ ;  
 $w_{\text{night}} = \log 2.5 - \log \frac{0.5}{0.8} = 0.916 + 0.470 = 1.386$ .  $w_0 = \log \frac{1}{999} - 0.811 - 1.153 - 0.470 = -9.340$ .  
. All flags:  $-9.340 + 7.781 = -1.560$ ,  $\sigma(-1.560) = 0.174$  ✓ — the same 210/1,209 as in the lecture.

3. Missing: drop the IP factor entirely (it sums to 1 over both values), odds  $6 \cdot 2.5/999 = 15/999$ ,  $P \approx 1.5\%$ .  
Plugging in  $x_{\text{IP}} = 0$  instead claims "domestic IP observed": it keeps the factor  $0.3/0.95$  (hidden in  $w_0$ ), odds  $4.74/999$ ,  $P \approx 0.47\%$  — a third of the honest answer.

4. Logistic regression models only  $p(y \mid x)$ ; it has no  $p(x_{\text{IP}} \mid y)$  to sum over. Its  $w_0$  is not split into per-attribute pieces, so there is nothing to remove — it must be given some value of  $x_{\text{IP}}$  or be refitted without the feature.

## Exercise 4.C.2: Counting Twice, in Code

Complete the three marked lines (numpy only), then run the script.

```
import numpy as np
rng = np.random.default_rng(42)
n = 5000
y = (rng.random(n) < 0.3).astype(int)          # 1 = fraud
flag = lambda pf, ph: (rng.random(n) < np.where(y == 1, pf, ph)).astype(int)
device = flag(0.6, 0.1)
copy = np.where(rng.random(n) < 0.95, device, 1 - device)  # a near-copy
night = flag(0.5, 0.2)
X = np.column_stack([device, copy, night])

def fit_nb(X, y, alpha=1.0):
    prior = ...                # (a)  $P(y = 1)$ 
    th1 = ...                   # (b) Laplace-smoothed  $P(x_j = 1 \mid y = 1)$ 
    th0 = ...                   # and  $P(x_j = 1 \mid y = 0)$ 
    w = np.log(th1 / th0) - np.log((1 - th1) / (1 - th0))
    w0 = np.log(prior / (1 - prior)) + np.log((1 - th1) / (1 - th0)).sum()
    return w0, w

def fit_lr(X, y, steps=5000, lr=0.5):  # gradient ascent, as in Lecture 03
    Xb = np.column_stack([np.ones(len(X)), X]); w = np.zeros(Xb.shape[1])
    for i in range(steps):
```

1. Print  $w_0$ ,  $w$  and  $P(\text{fraud} \mid x = (1, 1, 1))$  for both models.
2. Compare the weights of device and copy. Which model counts the device twice, and which shares the credit?
3. Which probability is closer to the fraction of frauds among the training rows with  $x = (1, 1, 1)$ ?

## Exercise 4.C.2: Counting Twice, in Code — Solution

# Exercise 4.C.2: Counting Twice, in Code — Solution

```
prior = y.mean()
th1 = (X[y == 1].sum(axis=0) + alpha) / ((y == 1).sum() + 2 * alpha)
th0 = (X[y == 0].sum(axis=0) + alpha) / ((y == 0).sum() + 2 * alpha)
```

1. (numpy 2.x, seed 42)

| model               | $w_0$ | $w_{\text{device}}$ | $w_{\text{copy}}$ | $w_{\text{night}}$ | $P(\text{fraud} \mid 1, 1, 1)$ |
|---------------------|-------|---------------------|-------------------|--------------------|--------------------------------|
| naive Bayes         | -2.80 | 2.60                | 2.17              | 1.35               | 0.965                          |
| logistic regression | -2.10 | 2.75                | -0.13             | 1.40               | 0.873                          |

2. Naive Bayes gives the copy almost the device's weight: the same evidence enters twice. Logistic regression fits all weights jointly, sees that the copy adds nothing once the device is known, and drives its weight to about 0.

3. Among the 486 training rows with  $x = (1, 1, 1)$ , 88.5% are fraud; the true posterior of the generating model is 0.865. Logistic regression (0.873) is close; naive Bayes (0.965) is **over-confident** — the double counting of Lecture 04, measured.